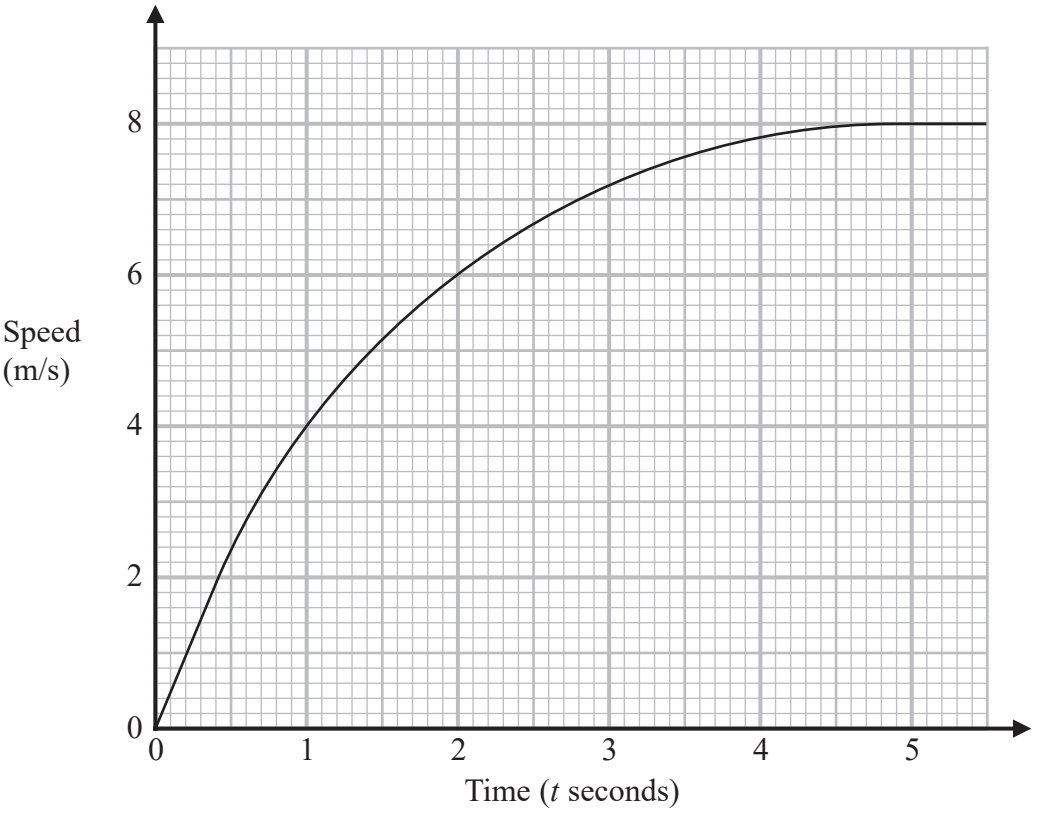
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

21 Here is a speed-time graph showing the speed, in metres per second, of an object t seconds after it started to move from rest.



(a) Using 3 trapeziums of equal width, work out an estimate for the area under the graph between $t = 1$ and $t = 4$

Read speeds at t=1,2,3,4: 4, 6, 7.2, 7.8.

Three trapeziums, each width 1 s:

A ≈ (1/2)[(4+6)+(6+7.2)+(7.2+7.8)]

= (1/2)(38.2) = 19.1

1 mark one trapezium

1 mark all three trapeziums

1 mark 19.1

19.1

(3)

(b) What does this area represent?

Area under speed–time graph = distance travelled.

(1)

1 mark distance travelled in metres

(Total for Question 21 is 4 marks)